

INTRODUCTION TO CCTV SYSTEMS



Closed Circuit Television (CCTV), or video surveillance, is a crucial part of modern security systems. Unlike public broadcasting, CCTV footage is sent to a specific, secure audience. Originally developed in 1942 to monitor V2 rocket launches, CCTV technology has advanced dramatically since then. Today, whether footage is stored locally or streamed to cloud services, CCTV systems are central to security strategies, offering real-time alerts to potential breaches.



However, the features that make CCTV systems so valuable-constant monitoring of sensitive locations and assets-also make them appealing targets for cybercriminals. These attackers exploit vulnerabilities such as default passwords, outdated firmware, and poorly configured networks to gain unauthorized access. This can lead to data breaches, interruptions in surveillance, or manipulation of footage. The consequences of such breaches are severe, including privacy violations, security failures, potential operational sabotage, and a significant loss of trust in security measures. This whitepaper explores various aspects of CCTV technology-from analog and IP cameras to advanced networked systems-and identifies the vulnerabilities in each. By examining recent cyberattacks and their impacts, this document aims to provide IT professionals with the knowledge needed

to implement robust security frameworks and effectively mitigate potential threats, ensuring the integrity and reliability of their surveillance systems.



UNDERSTANDING CCTV TECHNOLOGIES

The deployment of CCTV systems involves a variety of technologies and configurations, each with specific functionalities and vulnerabilities. This section explores the primary types of cameras, recording devices, and management systems integral to modern CCTV networks.

TYPES OF CCTV CAMERAS

CCTV cameras are broadly categorized into two types based on their data transmission and processing technologies:

1. **Analog Cameras**: These cameras operate by transmitting analog video signals directly to a Digital Video Recorder (DVR) through coaxial cables. The DVR then converts these analog signals into digital format for storage and playback. Analog cameras are less susceptible to cyber threats due to their lack of direct network connection, making them suitable for environments where basic surveillance without remote access suffices.

2. **IP Cameras (Internet Protocol Cameras)**

IP cameras, also known as network or digital cameras, capture and process video footage digitally within the camera itself and then transmit it over a network to a Network Video Recorder (NVR) or directly to cloud-based

storage. This type of camera offers higher resolution and better-quality images and facilitates remote monitoring and control. However, their connectivity to the network exposes them to potential cyber-attacks.

Each type of camera uses different methods for data transmission. Analog cameras rely on coaxial cables, while IP cameras can utilize Ethernet (Cat 5 or 6 cables), Wi-Fi, or even fiber optic cables if supported by the device. Choosing between analog and IP cameras often depends on specific security needs, budget, and the desired complexity of the installation.

RECORDING SOLUTIONS

CCTV systems typically employ one of two types of recording solutions:

1. **DVR (Digital Video Recorder)**: DVRs are designed for use with analog cameras. They receive analog signals which are then converted into digital before storage. DVRs require a direct connection to each camera via coaxial cables, which can limit the flexibility of camera placement and contribute to a more labor-intensive installation process.
2. **NVR (Network Video Recorder)**: NVRs are used with IP cameras and are more adaptable than DVRs due to their ability to receive and store digital video data sent over a network. This allows for more flexible installation options, as cameras can be positioned at considerable distances from the recorder without significant signal degradation, provided that network connectivity is maintained. Both DVR and NVR systems have distinct security profiles. DVRs are generally more secure from network attacks due to their lack of internet connectivity, while NVRs require robust cybersecurity measures, including strong passwords, network encryption, and frequent software updates to protect against unauthorized access.

BASIC THINGS TO CONSIDER BEFORE INSTALLING CCTV

1. **Coverage**: You will have to determine which areas are most vulnerable and need to be covered. The CCTV installation firm will send an analyst to help figure out the best possible location to mount cameras and

get optimal coverage. You can also let the professional know if there is any specific area you need the camera to cover.

2. **Types of CCTV Cameras:** **Wireless CCTV cameras** are the most commonly used nowadays. They eliminate the need to install wires. Moreover, these cameras can be easily controlled. But not all CCTV cameras are the same. The traditional directional camera, also called a **bullet camera**, is generally fixed and points only in one direction. On the other hand, **dome cameras**, commonly seen in malls and corporate offices, provide 360- degree coverage around them. Cameras need to be chosen based on the kind of coverage you need and the type of quality they can provide.

3. **Locations:** This point is related to what we discussed earlier regarding coverage. You should not forget about areas that are potentially vulnerable. This also means using different types of cameras based on their location. The camera that you place outside your garage door will not be the same as an indoor camera. Which brings us to the next point – do not ignore indoor coverage. Thieves and burglars are getting better at avoiding outdoor cameras and even total area coverage can be defeated by masked intruders. What these miscreants do not expect is the indoor camera. Placing a low-light capable dome camera indoors can save you a lot of trouble down the line.

4. **Declaration and Legality:** The law requires that you let people know that they are being recorded. This is important especially for public areas like a store or office. In both cases, signs stating that there are CCTV cameras installed will serve your purpose. For a home CCTV installation, you must notify any workers who come to your home. If you want to monitor your maid (or visitors outside the door), they must be informed verbally. Another important thing to note is that it is absolutely illegal to install CCTV cameras in private areas like bathrooms, dressing room, rest room or while in any area where the person has a reasonable expectation of privacy You can go to jail for such an offense.

5. **Permissions**: Depending on where you will be doing a CCTV installation, you may need to get official permission from relevant authorities. Even when it is not legally required, it is a form of courtesy. If you are installing CCTV cameras outside your home or shop, chances are that they cover a portion of a neighbor's home or store. Do let them know and change the angle, if they are uncomfortable. Most residential colonies require that you inform the local civic society or body (e.g. RWA) Resident welfare association, is a Non- governmental organization that represents the interests of the residents of a specific urban or suburban locality before installation. Home CCTV installation in apartment blocks usually requires permission too.

6. **Technical Aspects**: You will need to figure out some technical details about your security system. First, ensure that your cameras can record HD videos (high definition). Next, consider the additional equipment you need. Most CCTV installation is accompanied by their recording and storage hardware. Some also offer the option of cloud storage, which means storing the recorded video online. Local storage is done on a hard disk, just like on a computer or laptop. The difference between the hard disk of a camera system and your laptop is that the hard disk of a camera is of higher capacity and capable of running continuously for a much longer period. they also need to be replaced from time to time. Hence, you must also decide the duration for which you want to store any recorded video. You can choose anything from one day to three months or maybe even more if you have the storage capacity. Finally, opt for a system that offers remote control and access. This will let you monitor and control the CCTV camera even when you are traveling.

OTHER COMPONENT OF CCTV SYSTEM

1. VIDEO RECORDER
2. CAMERAS
3. MICROPHONE.
4. DISPLAY UNITE
5. MOUSE

6. HDD (HARD DISK DRIVE)
7. CENTRAL POWER PLITTER
8. POWER ADAPTOR (12V)
9. SWITCH (POE SWITCH)
10. ROUTER
11. POWER CABLE (RJ11 CONNECTOR)
12. VIDEO BALLON (CABLE REAPETER)
13. BNC CONNECTORS (BRITISH NAVAL CONNECTOR)
14. HDM I CABLE (HIGH-DEFINITION MULTIMEDIA INTERFACE)
15. RJ 45 CONNECTOR

TYPES OF CCTV CABLES

1. COAXIAL CABLE
2. RG 6
3. RG59 COAXIAL CABLE CARRIES BOTH VIDEO AND POWER
4. PLUG AND PLAY/ VIDEO AND POWER CABLE
5. UTP (Unshielded twisted pair Cable Cat5e/6) (NETWORK CABLE)
6. FIBER OPTIC CABLE

ABBREVIATIONS

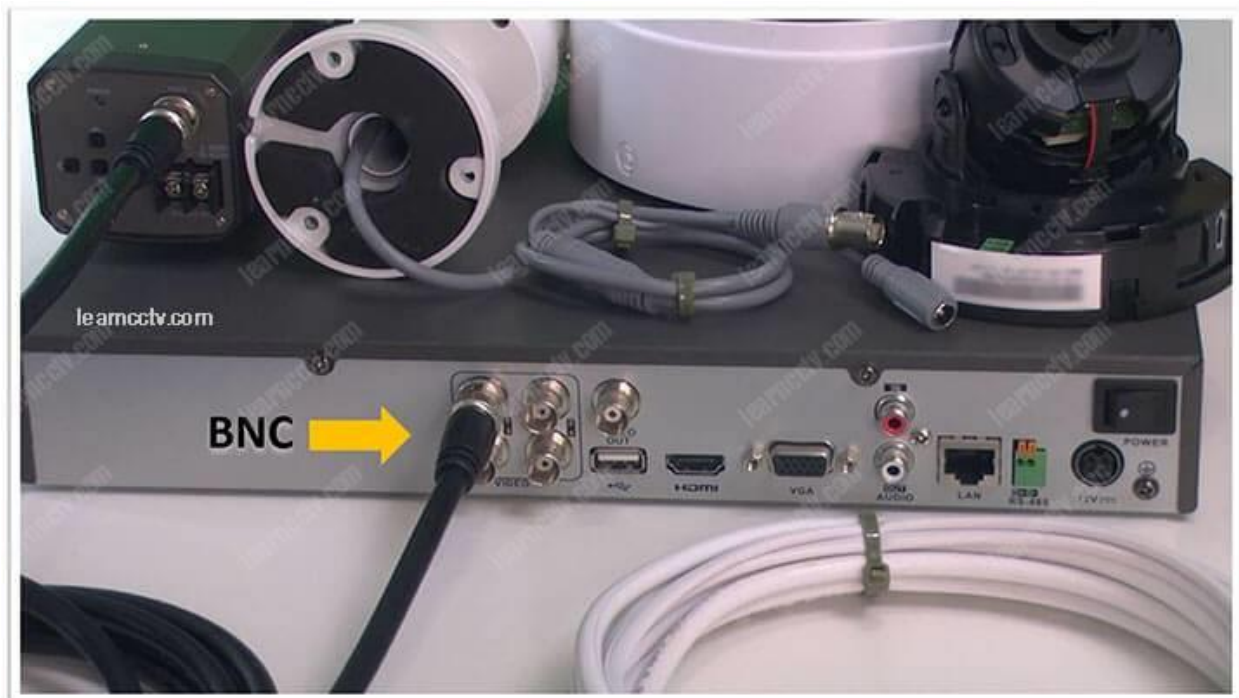
1. VCR = VIDEO CASSETTE RECORDER
2. DVR = DIGITAL VIDEO RECORDER
3. HD = HIGH DEFINITION
4. AHD = ADVANCE HIGH DEFINATION
5. HDD = HARD DISK DRIVE
6. BNC = BRITISH NAVEL CONNECTOR
7. HDMI = HIGH DEFINITION MULTIMEDIA INTERFACE
8. POE = POWER OVER ETHERNET
9. RJ = REGISTER JACK
10. RG = REDIO GUIDE OR GRADE
11. PTZ = PAN TILT ZOOM
12. VGA = VIDEO GRAPHICS INTERFACE OR ADAPTOR

13. IP = INTERNET PROTOCOL
14. DC = DIRECT CURRENT
15. CMOS = COMPLEMENTARY METAL OXIDE SEMICONDUCTOR
16. BIOS = BASIC INPUT OUTPUT SYSTEM
17. RAID = REDUNDANT ARRAY OF INDEPENDENT DISK

3 MAJOR COMPONENT OF CCTV SYSTEM

1. CAMERA
2. VIDEO RECORDER
3. DISPLAY UNITE

VIDEO RECORDER



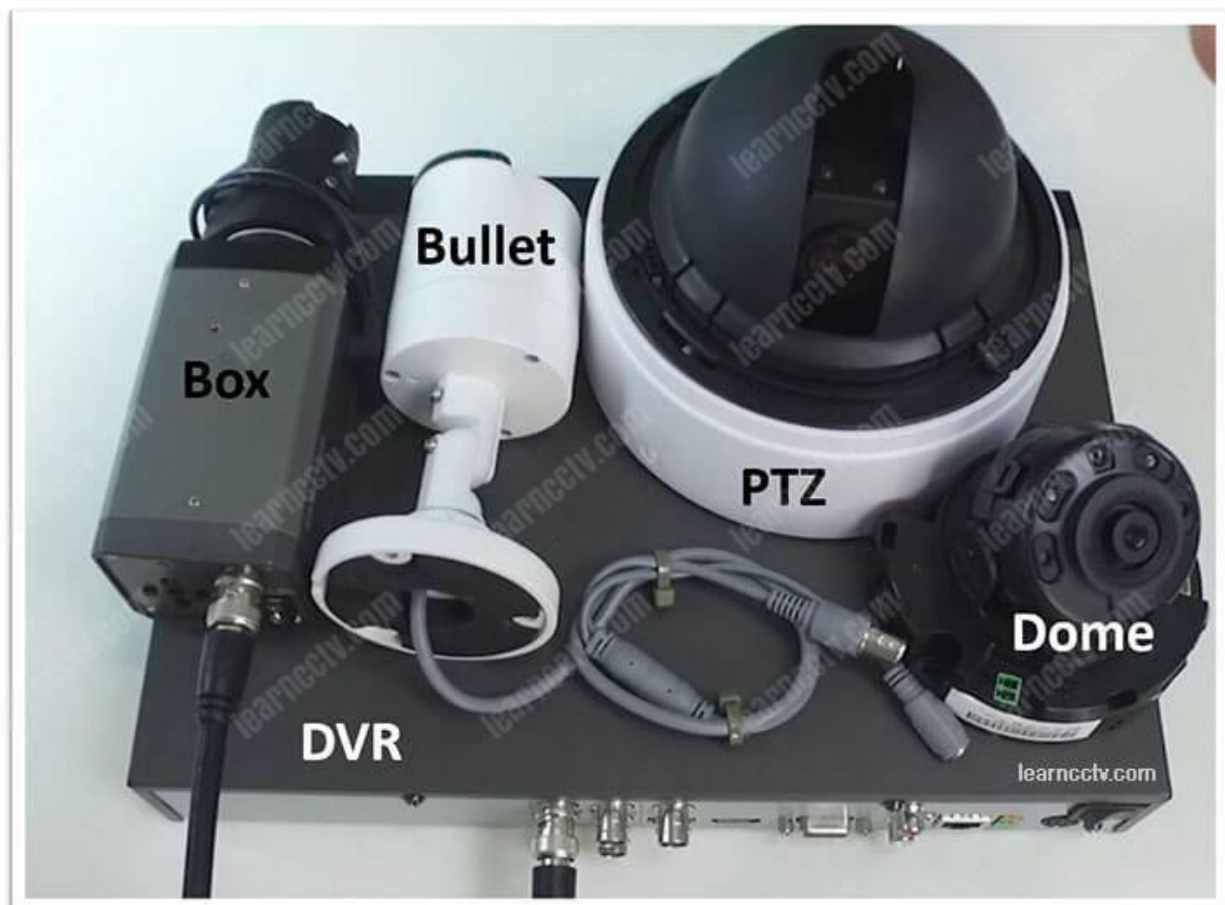
A video recorder is an electronic device that records video and sound to a disk drive, USB flash drive, SD memory card, SSD or other local networked mass storage device.

TYPES OF VIDEO RECORDER

It is possible to connect several analog cameras to a digital recorder (DVR), just get the equipment with an adequate number of channels. The following image shows **4 analog cameras** connected to a DVR that in turn is connected to an analog monitor that displays the images. The system uses coaxial cables designed for CCTV.



Take a look at the diagram, the **first block** shows the analog cameras that are responsible for generating the video, the **second block** has the DVR that converts the signals from analog to digital to record and send the video stream to be displayed in the **third block** where the analog monitor is located.



purpose of this image is only to illustrate the simplicity of the concept of



drive to record video. Therefore, DVRs are also called hard disk recorders. It is a computer system for image calculation, storage, and processing. It has the functions of recording, remote monitoring, and control of image/voice and dynamic frame for a longer time. The DVR integrates five functions, such as the video recorder, image splitter, PTZ lens control, alarm control, and network transmission. Its provides various interface, including video input/output, audio, storage expansion, VGA, HDMI, RS485, power DC 12v and network interfaces. DVR systems are considered the older legacy options. Unlike an NVR system, DVR systems work with Analog cameras, which can process video footage. the analog cameras stream raw video footage to recorder via coaxial cables while the recorder processes the video footage. The Analog Camera cannot process video data from the source

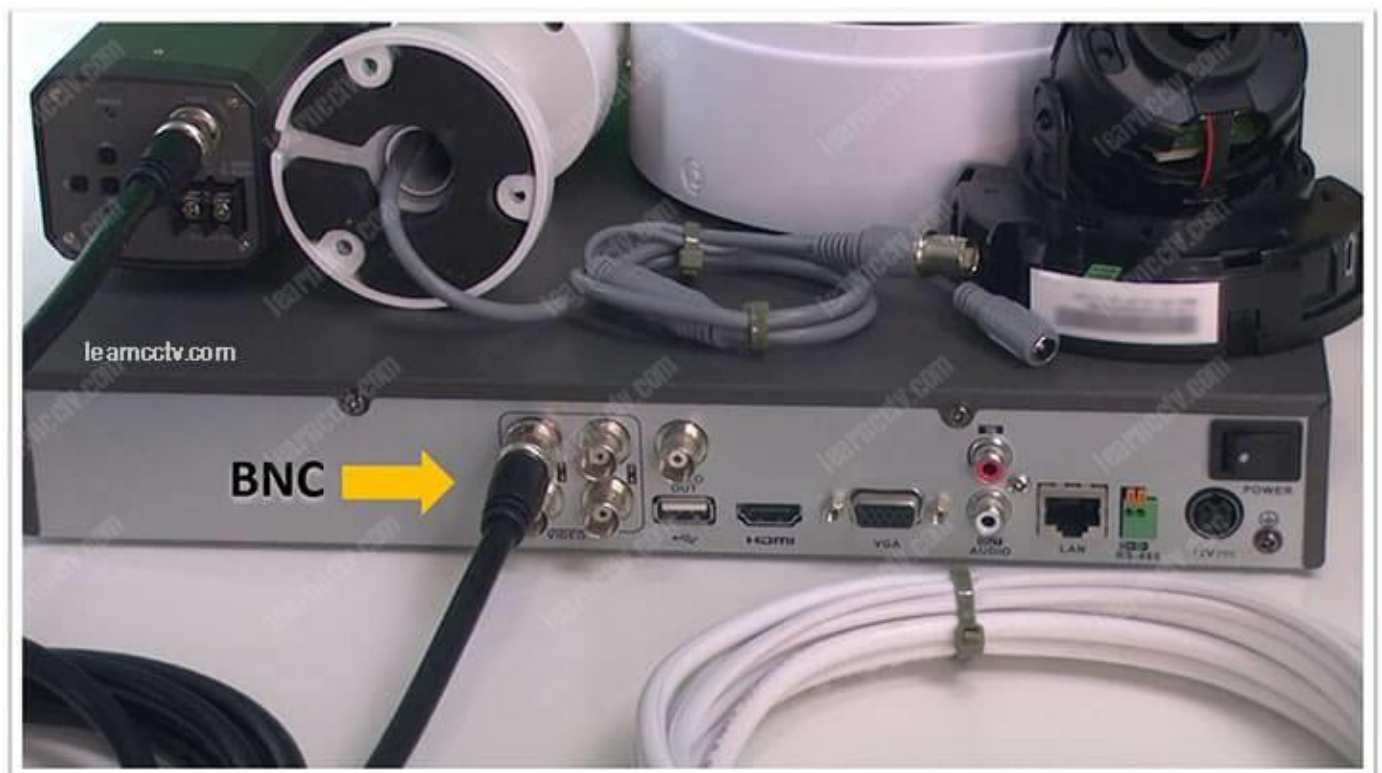
2. **NVR**: Network video recorder. Is a network recorder system with an IP address uses a twisted pair cable and a registered jack connector Connected via internet and can be access remotely? NVR and DVR perform the same functions they record video footage but The NVR systems has more advanced technology and therefore offer higher image/video quality among other benefits. NVR systems record and store video footage directly from the network with an advanced type of camera called IP cameras that can actually capture and process video and audio data themselves. through Ethernet cable or wireless via an existing WIFI network.



An analog camera captures images using a lens, image sensor and the internal circuitry system generates the analog electrical signals that are sent through coaxial cables to an external recorder (known as DVR) that converts analog signals to digital format for local recording and remote access. One of the cameras has the coaxial cable connected to the DVR so that it can send the electric signals to the DVR for digitization and recording.

The purpose of this image is only to illustrate the simplicity of the concept of connecting analog cameras to a digital recorder, it is also necessary to power all devices with a power supply.

The next image shows the back of a DVR, note that there are 4 inputs for the cameras and that the connection of the coaxial cable to the DVR is made by a BNC connector.



There are different types of coaxial cables on the market, in the picture above it is possible to notice that the black cable is thicker than the white one, but the last one has a parallel cable to carry the power to the camera (this type of cable is known as Siamese's /cable).



If a cable without parallel power cord is used (such as the black cable shown in the picture), it will be necessary to use a separate power cable, which may make the project a bit more expensive, but in some cases this solution is more professional, especially when the cameras are installed far from the DVR and the parallel cable is not good enough to carry the power.



In day-to-day situations for small projects, it is rare to find professional analog cameras of this type due to cost-effectiveness, especially when it is

better to use modern IP cameras rather than expensive professional analog cameras. If the design is small and your concern is cost, it is currently more common to use traditional analog cameras with siamese cables and power connectors.

TYPES OF DVR

- a. **Embedded DVR**: This is basically standalone device which works on its own, without a PC but obviously it can be connected to the local network and can access it using a PC browser on the same network. The operating system and DVR software are on the Chip. Called a pre-install software Though it does have a hard drive to record images, standalone normally have several internal Hard drive connections for extended recording. this system lessens the chances of virus and hackers.
- b. **PC- BASED DVR**: It is a PC with DVR Board, receives the video from the cameras and working in conjunction with the DVR software provides all the important video functions its operating system is mainly windows and can record surveillance footage. it can also browse and use other PC applications too. The recording equipment combined with a PC, A tower or rack mount cabinet contain a motherboard, a network card, a video board, a hard drive, and memory, plus a DVR capture card with video inputs for Analog cameras, along with Ethernet ports for Ip cameras. The DVR card is meant for recording in the computer.
- c. **HYBRID DVR**: Can record both Analog cameras and ip cameras to the same DVR. This allowed a mixture of regular Analog camera in low activity Areas and high definition (HD) ip cameras in high-risk areas for you to use a security camera system that incorporate both Analog and IP cameras using a Hybrid DVR.

TYPES OF NVR

1. **POE - NVR**: It only supports cameras that comply with the POE standard. The POE port on NVR is where these cameras are linked. A separate adaptor is not required for these recorders. The cameras and NVR can be connected using a CAT 5e /6 cable. This will enable data, audio, and video transmission over a single wire while also powering the camera.
2. **NON – POE NVR**. These NVR lack ports for connecting the camera to the rear panel. You require a POE switch or external power source to connect to the network camera. Most NON – POE NVRs have integrated scan capabilities that can find the network cameras. NVR can be set up with or without a power over Ethernet (POE) or non – POE switch. The cameras are connected to the POE port on the NVR, so no additional power adapters are required.

ADVANTAGE OF NVR SYSTEM

1. Record video and audio
2. Better image quality
3. More system flexibility
4. Better coverage, one camera has multiple lenses.
5. Can be wired or wireless
6. Only needs one cable for video, audio and power
7. Can be used to identify faces, license plates, due to higher image quality.

DISADVANTAGE NVR SYSTEM

1. NVR is more expensive
2. Software can be harder to use

ADVANTAGE OF DVR SYSTEM

1. Less expensive than NVR systems.

2. Simplicity, easier to set up and operate than NVR systems.
3. Image quality is improving (for systems using HD over coaxial technology).

DISADVANTGE DVR SYSTEM

1. Lower frame rates and lower image quality than NVR system.
2. Requires extra cables and wires, more complex installation process.
3. Separate power source required.
4. Less coverage, it takes multiple analog cameras to cover the same area as one IP camera.
5. Only good for observing.

CHANNELS

Are numbers of ports on the recorder that can accommodate securities cameras. DVR channels are really quite simple, since the work with analog CCTV systems, each security camera requires its own cable which must be plugged into its own port on the DVR. Each port corresponds to a channel. Eg A 4 channel DVR handle 4 security cameras, 8 channels DVR handle 8 security cameras while 16 channels DVR handles 16 security cameras. The numbers of cameras cannot really be separated from the number of channels the device can handle. DVR can only support as many security cameras, as it has channels ports on the back, it's a bit different with an NVR. Power over Ethernet, it is actually possible on some models to connect more security cameras than the number of ports on the back of the NVR, using a POE switch. A POE switch use more than one security camera into a single port, supplying power and combining both signals into a single Ethernet cable runs back to the NVR port. In most cases, you can easily double the number of cameras available to your NVR. You are not limited by the number of ports on the back of the DVR, you are limited by the channels dictated by the NVR firmware. your NVR might have 8 ports but be able to accommodate 16 channels, its simply recognizing the two multiple signals on a single cable as the two distinct security cameras they are on the other hand, it might be limited to merely 8 channels with those 8 ports. It depends on the exact model you selected for your CCTV system.

The exact number of channels you need depends on how many security cameras you have, and how many you foresee needing to install to proper coverage. Smaller businesses will likely be fine with an 8 or 16 channels DVR or NVR, but larger facilities (such as warehouses) will probably need a 32 to 64 channels or more to accommodate the necessary numbers of security cameras. In that case you will likely need an NVR and IP system to accommodate the sheer scale of the system. You might also be able to cut down on the number of cameras installed with PTZ cameras, for example. If you're just getting started with a few cameras but plan on expanding your system at any time in the foreseeable future, you might as well go with a 16 or 32 channels device and be prepared for adding cameras when the time comes.

CAMERA

A camera is a device for recording visual images in form of photographs, films, or video signals. Cameras are measure in pixel, pixel is the smallest dots or blocks that make up images in graphics shows. All images we see in computer monitor or televisions are made up of small dots. The higher the number of dots, the clearer the image. Example is the old type of TV stations that blinks black and white. Megapixel simply means one million pixel. Because pixels are small dots that forms an image in a picture, there is need for high number of dots to be able to get a clear imagery. It is the total numbers of this dots in a particular space that forms the resolution of a camera called pixel per inch (ppi). A camera with 720p (pixel) has lower quality imagery compared with a camera that has 1.3 mp (mega pixel) the ones with 3mp has higher image quality compared with those with 2mp and in that other. It will also be good to note that a screen with wide space needs more dots to display a good-looking image. So, the number of pixels (dots) in a space makes resolution of that device. A camera with 4mp has four million pixels in its display space. this term comes from the words (picture element) is the smallest portion of an image or display which a computer is capable of printing or displaying. This printed images, are created by several little DOTS. Each pixel has RGB (red, green, blue) color components. The brightness of each component is increased or decreased

to produce the variance of colors you see on the screen. The number of pixel determine the clarity of your video, the maximum number of pixel that can give you a clearer video with a D1(is a resolution standard for TV and video) Full D1 represents 720 x 480 pixels. with wide angle lens, the pixels spread apart very quickly as you move away from the cameras and fewer pixels mean less detail is captured. the pixels' cameras lens is measured in mm.

WHAT IS A PIXEL?

The word "pixel" means a picture element. Every photograph, in digital form, is made up of pixels. They are the smallest unit of information that makes up a picture. Usually round or square, they are typically arranged in a 2-dimensional grid The number of pixels in an image is sometimes called the resolution as the width and the height, for example a monitor resolution of 1280×1024. This means there are 1280 pixels from one side to the other, and 1024 from top to bottom. Another conversion is to express the number of pixels as a single number, like a 5-megapixel camera (a megapixel is a million pixels). This means the pixels along the width multiplied by the pixels along the height of the image taken by the camera equals 5 million pixels. In the case of our 1280×1024 monitors, it could also be expressed as $1280 \times 1024 = 1,310,720$, or 1.3 megapixels.

COLOR INFORMATION

Each pixel stores color information for image It usually stores in 3 components, known as RGB (Red, Green, Blue)

The number of distinct colors that can be represented by a pixel depends on the amount of information stored for each pixel. Information is stored as bits. the more bits per pixel (bpp) that are stored, the more colors a pixel can represent. For example, if only a single bit of information is stored for a pixel, then it can be "on" or "off" black or white. The actual number of bits used to represent the color of a single pixel is known as color depth, or bit depth

An approximate guide to the optical range of different lens cameras, the distance at which a number plate or a person can be identified using standard definition. For HD1080p systems.

- | | | |
|----|------------|------------|
| a. | 2.8mm lens | 1.5 meters |
| b. | 6mm lens | 3 meters |
| c. | 6mm lens | 5 meters |
| d. | 9mm lens | 6 meters |
| e. | 12mm lens | 7 meters |
| f. | 22mm lens | 12 meters |
| g. | 60mm lens | 35 meters |

PART OF A CAMERA

1. Focusing screen:
2. Image sensor: it converting light in to electronic signals.
3. Lenses: focusing on clarity and details.
4. Infrared: illuminations shedding light on the dark
5. Mounting and covers. The foundation of camera

TYPES OF CCTV CAMARAS

1. **ANALOG CAMERA**: An analog camera is a traditional camera used in CCTV systems. Its sends video over cables to the DVR. Analog camera has overall lower quality than IP cameras but perform better in low light conditions. The have more limited site ranges and don't offer the zoom in clarity of an IP cameras. If you zoom in on the analog images, you're going to get a grainier, degraded picture. You're not going to recognized the perpetrators face by zooming in. the cameras are limited to resolutions of the NTSC/ (National television standard committee) standard of 720 x 480 pixels. The analog camera resolutions range from 420 to 700, which at the high end can produce sharp images. Analog cameras can send video over twisted pair cable up to 1.5 kilometers away and 300 meters away over coaxial cable. But analog transmissions loss clarity with increased distance and when the signals is converted from one format to another. Its require three separate cables e.g. power, audio, video.

2. **IP CAMERA**: An internet camera, or IP cameras, is a type of digital video cameras that receives data and sends image via an IP network. Some IP cameras require support of a central network to handle the recording, video and alarm management. Other are able to operate in a decentralized manner with no NVR needed, as the cameras is able to record directly to any local or remote storage media. IP cameras used for home security and a live video stream to a companion app on the user's phone, connect to the internet through WIFI, broadband, or Ethernet cable. One of the most popular abilities is that you can view their cameras footage via a mobile app or other application software. Many cameras offer the features of a wide-angle lens around 140 degrees, or pan/tilt up to 360 degrees' horizontal, 90 degrees vertical, low light or night vision, and motion detection when an event occurs, such as detected motion, Rs can receive alarms and notifications via an APP. Video clip can be stored in a local device such as a micro – SD or through a cloud service.

SIZE OF CCTV CAMERAS

1. MINI CAMERA
2. MICRO CAMERA (SPY CAMERA)
3. MEDIUM CAMERA

The micro and the mini cameras have a short lance that cannot go far and the normally don't have inferred. The work of the inferred is to give light to the camera whenever is dark.

Advantages of CCTV

Security is an important factor in both the home and workplace. Anything that increases security measures has obvious advantages. Some of these pros include:

1. Increased safety and security.
2. 24/7 surveillance.
3. Crime deterrent with visible cameras.
4. Evidence to any criminal activities caught on tape.
5. Set up remote access.

6. Discounted insurance

Whether you set up a CCTV to help protect your family or employees, it's well worth the investment. But there are still some issues to be careful about.

Disadvantages of CCTV

One of the biggest disadvantages of CCTV use deals with privacy, especially when used in the workplace. While it may be there to help keep employees and customers safe, they may object to being filmed under constant surveillance.

CATEGORIES OF CCTV CAMERA

1. **Dome CCTV Cameras**



A dome CCTV camera gets its name from the dome-shaped casing that the camera sits in. Whilst these are relatively discreet CCTV cameras in appearance, this doesn't stop them from deterring criminals. This is because the dome casing makes it really difficult for people to see which direction the camera is pointing. This creates an air of uncertainty for potential thieves or vandals approaching from all directions.

Main benefits

- a. Unobtrusive design means it's easy to fit
 - b. Suitable for indoor and outdoor use
 - c. Vandal resistant dome means it is harder to interfere with the camera
 - d. 360-degree rotation of the camera so you can cover all angles
- Ideal uses

This is the best type of CCTV camera for shops, restaurants, casinos and hotels because the design allows you to monitor a wider angle at your premises. Many also consider this option to be the best indoor CCTV camera as it can be placed on a strategic point of the ceiling to cover all angles of the room.

2. **Bullet CCTV Cameras**



Overview: Bullet CCTV cameras have an iconic design that is highly visible. They are cylindrical in shape and are capable of observing long distances. Bullet cameras are most commonly placed outdoors so their casings are made resistant to water, dust and dirt.

Main benefits

- a. Highly visible so acts as a deterrent.
- b. Resistant to dirt in challenging environments.
- c. Provides surveillance over long distances.
- d. Casing also protects against glare and rain.

Ideal uses

Bullet cameras can be used to good effect in a range of industries such as manufacturing, property management and farming. They not only record good quality pictures but also send out a message to passers-by that your premises is under surveillance. That's why they have traditionally offered the best CCTV for outside operation.

3. **C-Mount CCTV Cameras**



C-Mount CCTV cameras are the ideal option for those who may wish to adapt the field of vision. They are equipped with detachable lenses that can be switched around to monitor varying distances. C-Mount cameras are typically bulky so, like bullet cameras, act as a visible deterrent to criminal activity.

Main benefits

- a. Special lenses can be fitted to the camera to extend its range beyond 40 feet.
- b. Highly visible appearance that deters criminals.
- c. Rugged camera casing can be added to protect the camera in cold environments.
- d. Weatherproof housing makes it ideal for outdoor use.

Ideal uses

This is the ideal camera for observing a stretch of road on the approach to your premises. It is therefore popular in the utilities and logistics industries. Those operating in the food and manufacturing industries may also elect for this option, due to the camera's reliability in extreme temperatures.

4. **PTZ Pan Tilt & Zoom Cameras**

Overview



With a PTZ (Pan Tilt & Zoom) camera, your security team can have complete control on what is recorded. At the touch of a button, the camera lens can pan left and right, tilt up and down or zoom in and out. It's the ideal choice if you have a security guard who is monitoring a live video feed on site.

Main benefits

- a. Optical zoom on these cameras means you can focus closely in on subjects.
- b. Pan and tilt feature provides 360-degree field of vision.
- c. Image resolution is usually impeccable so facial features can be distinguished.
- d. Security team have full control of recording and can react to live situations

Ideal uses

This is the best CCTV camera for remote viewing. It is really popular in large retail environments and is regarded as the best CCTV for business premises that have a specific valuable object to protect. For example, a museum could program the camera to pan to a particular entry or exit point when it detects movement. So even when it is not being controlled remotely, the camera can cover a much larger area and do the same job of several static cameras.

5. Day/Night CCTV Cameras



Overview

These cameras have been built specifically to operate effectively, regardless of how well lit their environment is. During both daytime and night-time, they will record clear video images that will not be obscured by differing light condition. This is achieved through their extra sensitive imaging chips.

Main benefits

- a. Clear recording in low light.
- b. They usually record in both color and black and white.
- c. Can function despite glare, direct sunlight, reflections and poor lighting.
- d. Casing protects the camera against the elements for outside surveillance

Ideal uses

There are various CCTV camera types suited to outside use but the extra sensitive imaging makes this one unique. It is therefore the ideal CCTV system for business premises that require round the clock CCTV monitoring in an outdoor environment with varying light conditions.

6. Infrared/Night Vision CCTV Camera



As the name suggests, these cameras are designed to operate optimally in pitch black conditions. They achieve this by using infrared technology. Infrared models are more expensive than day/night cameras and those

opting for this option usually do so because night recording is vital to the security of their business.

Main benefits

- a. Infrared LEDs allow clear recording even in pitch black environments.
- b. An infrared cut filter activates in the daytime for clear images in lighter conditions.
- c. Can capture images even in the presence of fog, dust or smoke.
- d. Records color imagery in the daytime and black and white at night.

Ideal uses

This is the best CCTV camera for night vision functionality. It's therefore the ideal choice for those who need ultra-reliable recording in complete darkness. For example, banks, farms and manufacturing plants.

7. Network/IP CCTV Cameras



These cameras share live footage across the internet so images can be easily accessed from anywhere on the globe. The bandwidth of the video is compressed, in order to make the online feed reliable. Archive footage is stored on network video recorders (NVRs) or on secure software for later access.

Main benefits

- a. Footage can be viewed online from anywhere in the world.
- b. Easily accessible archive footage.

- c. Easy setup as co axial cables and a computer station are not required.
- d. Low maintenance

Ideal uses

A network or IP CCTV camera is the ideal choice for business owners who are often away from site and on the move. Footage can be viewed from anywhere in the world, via a PC, laptop or mobile phone.

8. Wireless CCTV Cameras



Wireless CCTV cameras were created to minimize [HYPERLINK "https://www.businesswatchgroup.co.uk/security-services/cctv-systems/installation/"](https://www.businesswatchgroup.co.uk/security-services/cctv-systems/installation/)installation time. They also make the appearance of the camera much tidier, which may be an important factor for locations such as churches, museums or stately homes.

Main benefits

- a. Easier to install than your average CCTV system.
- b. Discreet and tidy appearance and a less obtrusive fitting.
- c. Images transmitted over internet so can be viewed anywhere.
- d. Secure storage and easy access of archive footage.

Ideal uses

If a spotless presentation is important in your industry, wireless CCTV cameras are a great security option. Their discreet appearance helps them

to blend into the interior. footage is transferred via the internet for convenient access.

9. **High Definition (HD) CCTV Cameras**



Overview

HD CCTV cameras offer an unrivalled picture quality that delivers high clarity images. Depending on your budget, they can deliver resolutions of 720p, all the way up to 4K. Therefore, there's no danger that the subject of recordings will ever appear grainy or distorted.

Main benefits

- a. High definition pictures capture subjects in perfect clarity
- b. Extra clarity can also be achieved with zoom function
- c. Available on most CCTV models such as domes and bullets
- d. Crystal clear imagery provides undeniable criminal identification.

Ideal uses

High-definition cameras are particularly use in environments where there is a high risk of theft. Businesses such as banks and casinos should opt for HD cameras, in order to ensure that visitors to their premises are captured in perfect clarity. The footage recorded will be able to provide undeniable identification in the event of a crime being committed.

Some cameras are capable of powering themselves with the same cable that runs to the NVR. This is rare. Another subsection of cameras that use batteries instead of a power cable of any kind. These too are rare; batteries need to be changed often if a camera system is constantly being used.

For all other security camera systems, the power is run through a power supply unite, the power supply connects the camera to a wall outlet or power regulator. You can have a single power supply or be able to power all the way up to 16 cameras on one power supply.